Nermine Moumen

Final-year Engineering Student in Industrial Chemistry

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EDUCATION

National Institute of Applied Sciences and Technology (INSAT) *2022–Present*

Industrial Chemical Engineering student, specialized in Process and Environment, with a strong foundation in electrochemical kinetics, electrochemical measurements and materials characterization. Interested in the experimental evaluation of metallic materials and the interpretation of their electrochemical behavior. Seeking an internship starting February 2027.

PROFESSIONAL EXPERIENCE

SEGULA Technologies – Electrochemical Battery Modeling Internship *06/2026–08/2026*

- Modeled a **Li-ion cell** (NMC811/Graphite-SiOx) using the Single Particle Model (SPM) implemented in MATLAB.
- Implemented the electrodes' **open-circuit potential (OCP) curves** and solved the PDEs with **FVM**.
- Optimized the mesh and diffusion coefficient values, reducing the RMSE error **from 202.13 mV to 40.5 mV**.
- Extracted and processed **experimental discharge data** from a database to validate MATLAB and COMSOL results.

Control Union – Quality Control Internship *08/2025–09/2025*

- Performed **laboratory analysis** of lubricants (viscosity, heavy metals by **ICP-OES**) following standardized procedures.
- Prepared and interpreted **analytical reports** to verify compliance with ASTM D5185 and **ISO 17025** standards.

ETAP – Water Treatment and Petrochemistry Internship *07/2024–08/2024*

- Performed physico-chemical and ionic analyses of water samples (pH, salinity, chlorides via ion chromatography).
- Studied water treatment methods **(coagulation-flocculation, ion exchange)** for contaminant and **corrosion** control.
- Performed organic geochemistry techniques **(GC, GC-MS, Rock-Eval pyrolysis)** for hydrocarbon **characterization**.

ADDITIONAL EXPERIENCE

Cascade – Online Chemistry Teacher (Part-time) *2024–Present*

- Provide individualized online chemistry tutoring for French Terminale students, covering exam-level concepts.

ACADEMIC PROJECTS

Electrochemical Treatment of an Industrial Effluent by Electroflocculation

- Carried out **electrochemical measurements** on aluminum **electrodes**: current-potential curves and **pulse polarography**.
- Calculated the **faradaic yield** with **Faraday's law**, linking electrode dissolution to the charge passed.
- Removal of Cd^{2+} and Zn^{2+} from a simulated effluent: **85% and 57%** in 20 min.

Molecularly Imprinted Polymers (MIPs) for Pharmaceutical Applications

- Synthesis and characterization of a sulpiride-selective MIP via radical polymerization, with **Soxhlet extraction**, **UV-Vis** spectrophotometry monitoring, and validation through controlled-release kinetics (240 min).
- Synthesis of a **core-shell MOF/polymer MIP** for targeted doxorubicin delivery, achieving 85% encapsulation efficiency.

PEM Fuel Cell and Electrolyzer Modeling with COMSOL and ANSYS Fluent

- Modeling of fuel cells (two-phase flow, **thermal effects**) at **imposed current** then at **imposed potential**.
- Modeling of a 3D PEM electrolyzer for **hydrogen production**: charge conservation, identification of the OER.

Prediction of Liquid–Liquid Equilibria and Cross-flow and Countercurrent Extraction for Ternary Systems

- Formulated the LLE thermodynamic system and solved it via Rachford-Rice and **arc-length continuation**.
- Reproduced 7 Mathematica notebooks proving equivalence of the **Hunter-Nash** and **McCabe-Thiele** methods.
- Reimplemented workflow in **Python**, MATLAB, and Mathematica (spline/fsolve, SciPy/mpltern, Interpolation/FindRoot).

TECHNICAL SKILLS

- **Lab Techniques:** Chromatography (HPLC, GC, TLC), Spectrometry (MS, ICP-OES), Spectroscopy (NMR, FTIR, UV-Vis), Diffraction (XRD), Microscopy (SEM, TEM), Electrochemistry (Cyclic voltammetry, Polarography).
- **Simulation:** Aspen Plus, Aspen HYSYS, COMSOL Multiphysics • **Programming:** *MATLAB, Mathematica, Python*
- **Data & Office:** Microsoft Office Suite, OpenLCA • **Standards:** ISO 9001, ISO 14001, ISO 45001, ISO 17025, ISO 14040

EXTRACURRICULAR ACTIVITIES

- **IEEE INSAT** – Chair of EMBS; Organizing Chair of TRSYP 3.0; Social Activities Chair of MENASYP.
- **THEATRO INSAT** – Member of the dance team.

LANGUAGES AND CERTIFICATES

Languages

English (Fluent), French (Fluent), Arabic (Native).

Certificates

Aspen HYSYS – MATLAB